The Sugar Vault

An Arduino controlled candy & chocolate dispensing machine

MAIN

Start Date: 5/18/2026
End Date: 6/28/2026
Software Used: Google Docs, SOLIDWORKS, draw.io, Arduino IDE, GitHub, Bambu Studio
Project GitHub: https://github.com/cchapman9x/the-sugar-vault

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Project Creator:

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Project Documentation Index:

- TheSugarVault_PARTS+BOM.pdf
- TheSugarVault_CODE_BREAKDOWN.pdf
- TheSugarVault_DESIGN_BREAKDOWN.pdf

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Project Mission Summary

The Sugar Vault is an Arduino-controlled candy and chocolate dispensing machine designed to dispense candy using a motorized wheel mechanism. The project uses an Arduino Nano, an A4988 stepper motor driver, a NEMA 17 stepper motor, a push button, an LED indicator, a buck converter, and custom 3D-printed mechanical parts.

I created The Sugar Vault for two main reasons. The first reason was to improve my engineering skills by building a complete physical project that combined electronics, programming, CAD, 3D printing, wiring, and mechanical troubleshooting. I wanted a project that was realistic for my current skill level but still challenging enough to teach me new concepts about microcontrollers, motor control, and system integration.

The second reason was more personal. I wanted to build something fun that could eventually be placed in the office where I work so that my coworkers and I could enjoy treats while doing our jobs. The Sugar Vault is not only an engineering project; it is also a creative project based on an older candy dispenser idea I made in high school. I have always wanted to build off of it, so I used this project as the perfect opportunity to do so.

The final version of The Sugar Vault uses a push button as the main input. When the button is pressed, the Arduino controls the A4988 stepper motor driver, which drives the NEMA 17 stepper motor. The motor rotates the dispensing wheel, allowing candy to drop from the storage tube and exit through the dispenser opening. An LED turns on during the dispensing cycle to show that the machine is active.

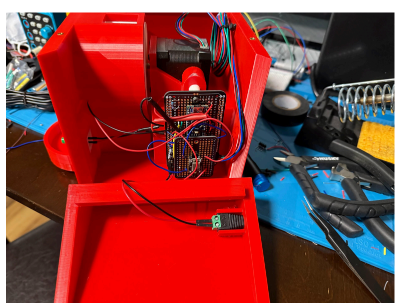
This project served as an introduction to mechatronics because it required the electrical, mechanical, and software parts of the system to work together.

Project Goals

The main goal of The Sugar Vault was to design and build a working Arduino-controlled candy dispenser from start to finish.

The **project goals** were:

- Design a candy dispenser that uses electronic control instead of manual operation
- Use an Arduino Nano as the main controller
- Control a NEMA 17 stepper motor using an A4988 stepper motor driver
- Use a push button as the user input
- Use an LED as a visual indicator during dispensing
- Power the system using an external wall power supply
- Use a buck converter to provide safe logic-level voltage for the Arduino
- Design and 3D print the mechanical parts needed for the dispenser
- Create a wiring diagram for the electronics system
- Test and troubleshoot the dispenser until it worked
- Document the finished project on GitHub

		
Figure 1	**Figure 2**	**Figure 3**

Final Report & Project Reflection

The final version of The Sugar Vault successfully dispenses candy using a push button-controlled stepper motor system. The project combines an Arduino Nano, A4988 stepper motor driver, NEMA 17 stepper motor, buck converter, external power supply, LED indicator, a clear acrylic tube as the hopper, and custom 3D-printed parts into one complete working prototype.

This project helped me practice several important engineering skills. I learned more about wiring electronic components together, using an external power supply, controlling a stepper motor, working with a motor driver, designing parts in CAD, preparing files for 3D printing, how to split power for different components, and troubleshooting a full physical system.

One of the most important lessons from this project was that a working engineering design is not only about the electronics or the code. The mechanical design matters just as much. Even when the circuit and code are correct, the machine still has to physically move without jamming, rubbing, or stalling. During testing, The Sugar Vault worked, but I noticed that the dispensing wheel could occasionally get stuck when too much candy weight pressed down on it. This appears to be a mechanical load and friction issue rather than a software issue. The motor and code can rotate the wheel, but the weight of the candy can sometimes create enough resistance to stall the mechanism.

Overall, The Sugar Vault was completed as a functional Arduino-based mechatronics project. It achieved the main goal of creating a motorized candy dispenser while also showing clear areas for future mechanical improvement.

Known Limitations & Future Improvements

The main limitation of the current design is the weight of the candy pressing down on the dispensing wheel. During testing, I noticed that the motor would sometimes stall after the button was pressed, preventing the wheel from turning. After investigating the issue, I concluded that too much candy weight was resting directly on the dispensing wheel, creating enough friction and resistance to prevent the motor from rotating it consistently. The shape of the candy may have also contributed to the wheel getting stuck, especially when combined with the weight of the candy pressing down on the mechanism.

In the future, I plan to address this issue by designing a slanted holding piece or **load-relief baffle** inside the hopper. This part would help prevent most of the candy weight from sitting directly on top of the dispensing wheel, allowing the wheel to rotate more reliably during dispensing.

Even with this limitation, the project still works as intended. However, the design could benefit from further revisions and improvements.

Future improvements I would like to implement include:

- Adding a load-relief baffle above the dispensing wheel
- Reducing friction between the dispensing wheel and the housing
- Improving the geometry of the wheel pockets
- Testing different candy sizes and shapes
- Improving reliability when the candy tube is filled with more weight
- Creating a cleaner internal wiring layout
- Designing a custom PCB for the electronics system
- Improving the exterior finish of the enclosure
- Using a different filament for the final version (PETG is a very stringy filament)

Reference Materials & Resources:

- https://roboticsbackend.com/arduino-input_pullup-pinmode/
- https://www.circuits-diy.com/control-led-with-push-button-arduino-tutorial/
- https://leecuriosity.com/arduino-nano-pinout-full-guide-for-beginners/
- https://www.youtube.com/watch?v=wcLeXXATCR4&t=810s
- https://forum.arduino.cc/t/how-to-drive-and-power-a-nema-17/1002058/4
- https://howtomechatronics.com/tutorials/arduino/stepper-motors-and-arduino-the-ultimate-guide/
- https://forum.arduino.cc/t/how-to-drive-and-power-a-nema-17/1002058
- https://www.youtube.com/watch?v=6YO_21WFFzI